

Homework

Topic: Limits, Continuity, Basic & Advanced Differentiation (Units 1-3)

Instructor: Lain

Part I: Formulas and Techniques

1. Limits & Continuity

- 0/0 (L'Hôpital's Rule)
- $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$ $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x} = 0$
- () **BOBO BOTN EATS DC**
 - Bottom heavy $\implies y = 0$
 - Top heavy $\implies$ None (∞)
 - Exponents same $\implies$ Divide Coefficients ()
- $f(c)$ $\lim_{x \rightarrow c} f(x)$ $\lim_{x \rightarrow c} f(x) = f(c)$

2. Derivative Rules

- $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ $f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$
- **Power Rule** () $\frac{d}{dx} [x^n] = nx^{n-1}$ ()
- **Product Rule** () $y' = f'g + fg'$ (+)
- **Quotient Rule** () $y' = \frac{\text{Low } d(\text{High}) - \text{High } d(\text{Low})}{(\text{Low})^2}$
- **Chain Rule** () $y' = f'(g(x)) \cdot g'(x)$ ()

3. Key Functions

- $\frac{d}{dx} [e^x] = e^x$
- $\frac{d}{dx} [\ln x] = \frac{1}{x}$
- - $\frac{d}{dx} [\sin x] = \cos x$
 - $\frac{d}{dx} [\cos x] = -\sin x$
 - $\frac{d}{dx} [\tan x] = \sec^2 x$
 - $\frac{d}{dx} [\sec x] = \sec x \tan x$
 - $\frac{d}{dx} [\cot x] = -\csc^2 x$
 - $\frac{d}{dx} [\csc x] = -\csc x \cot x$
 - (C)

- $\frac{d}{dx} [\arcsin x] = \frac{1}{\sqrt{1-x^2}}$
- $\frac{d}{dx} [\arctan x] = \frac{1}{1+x^2}$

4. Advanced Techniques

- **Implicit Differentiation** () y $\frac{dy}{dx}$
 xy
- **Inverse Functions** () $(f^{-1})'(b) = \frac{1}{f'(a)}$ ()
- **Tangent Lines** ()
 - $y - y_1 = m(x - x_1)$ $m = f'(x_1)$
 - **Horizontal Tangent** () $= 0$
 - **Vertical Tangent** () $= 0$

Part II: FRQ Templates

1 $x = c$ (Continuity)

1. $f(c) = [\text{value}]$
2. $\lim_{x \rightarrow c^-} f(x) = \lim_{x \rightarrow c^+} f(x) = [\text{value}]$ (Therefore, $\lim_{x \rightarrow c} f(x)$ exists)
3. Because $\lim_{x \rightarrow c} f(x) = f(c)$, the function $f(x)$ is continuous at $x = c$.

2 (Intermediate Value Theorem - IVT)

$$c \quad f(c) = k$$

"Because $f(x)$ is **continuous** on the closed interval $[a, b]$, and k is strictly between $f(a)$ and $f(b)$ (or write $f(a) < k < f(b)$), by the **Intermediate Value Theorem (IVT)**, there must exist a value c in (a, b) such that $f(c) = k$."

Part III: The Test

Directions: Solve each of the following problems. Calculators are strictly prohibited.

Section A: Multiple Choice Questions (AP Style)

1. Evaluate $\lim_{x \rightarrow 2^-} \frac{|x-2|}{x-2}$.
 (A) 1
 (B) -1

- (C) 0
(D) The limit does not exist

2. Evaluate $\lim_{x \rightarrow \infty} \frac{\sqrt{9x^2+2}}{4x+3}$.

- (A) $\frac{3}{4}$
(B) $\frac{9}{4}$
(C) $\frac{3}{2}$
(D) ∞

3. What is $\lim_{h \rightarrow 0} \frac{\cos(\pi+h) - \cos(\pi)}{h}$?

- (A) -1
(B) 0
(C) 1
(D) The limit does not exist

4. Evaluate $\lim_{x \rightarrow 0} \frac{\sin(4x)}{5x}$.

- (A) 0
(B) $\frac{4}{5}$
(C) $\frac{5}{4}$
(D) $\frac{4}{5}x$

5. Let $f(x)$ be defined as follows: $f(x) = \begin{cases} x^2 + ax, & \text{for } x \leq 1 \\ bx - 2, & \text{for } x > 1 \end{cases}$ If $f(x)$ is differentiable at $x = 1$, what are the values of a and b ?

- (A) $a = -5, b = -3$
(B) $a = -2, b = 0$
(C) $a = 1, b = 3$
(D) $a = -3, b = -4$

6. If $f(x) = \ln(3x^2 + 1)$, what is $f'(1)$?

- (A) $\frac{1}{4}$
(B) $\frac{3}{4}$
(C) $\frac{3}{2}$
(D) $\frac{1}{2}$

7. If $y = x^2 \sin(2x)$, then $\frac{dy}{dx} =$

- (A) $2x \cos(2x)$
(B) $4x \cos(2x)$
(C) $2x \sin(2x) + x^2 \cos(2x)$
(D) $2x \sin(2x) + 2x^2 \cos(2x)$

8. Let f be the function defined by $f(x) = x^3 + x - 2$. Let g be the inverse function of f . What is the value of $g'(0)$?

- (A) $\frac{1}{4}$
 (B) $\frac{1}{3}$
 (C) 1
 (D) 4

9. What is the slope of the line tangent to the curve $y^3 + 2x^2y = 8$ at the point $(2, 1)$?

- (A) $-\frac{4}{11}$
 (B) $-\frac{8}{11}$
 (C) $-\frac{4}{7}$
 (D) $\frac{4}{11}$

10. If $f(x) = e^{\tan x}$, then $f''(x) =$

- (A) $e^{\tan x} \sec^2 x$
 (B) $e^{\tan x} (\sec^4 x + 2 \sec^2 x \tan x)$
 (C) $e^{\tan x} \sec^2 x \tan x$
 (D) $e^{\tan x} (\sec^2 x + \tan x)$

Section B: Free Response Questions

**11. Let f and g be differentiable functions. The table below gives values of f , f' , g , and g' at selected values of x .

x	$f(x)$	$f'(x)$	$g(x)$	$g'(x)$
1	3	-2	2	4
2	5	1	3	-3
3	-1	4	1	2

(a) Let $h(x) = f(x) \cdot g(x)$. Find the exact value of $h'(1)$.

(b) Let $k(x) = \frac{f(x)}{g(x)}$. Find the exact value of $k'(3)$.

(c) Let $p(x) = f(g(x))$. Find the exact value of $p'(2)$.

**12. Consider the curve defined by the equation $x^2 + xy + y^2 = 27$.

(a) Write an expression for the slope of the curve at any point (x, y) . Show that $\frac{dy}{dx} = \frac{-2x-y}{x+2y}$.

(b) Find the coordinates of all points on the curve where the tangent line is vertical.

(c) Find the value of $\frac{d^2y}{dx^2}$ at the point $(3, 3)$.

**13. Let $f(x)$ be a function defined as $f(x) = xe^{2x}$.

(a) Find the equation of the line tangent to the graph of f at $x = 0$.

(b) Let $W(x) = f(x) - x - 2$. Use the Intermediate Value Theorem to justify that there exists at least one value c in the interval $(0, 1)$ such that $W(c) = 0$.